

Linlin Jia

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About Me

I am a postdoc at the Pattern Recognition Group, University of Bern, working with PD Dr. Kaspar Riesen. I work on graph and spatio-temporal machine learning. On the theory side I study graph spaces: graph kernels, graph edit distances, and the pre-image and generation problem. I also care about methods that carry across applied domains, and about turning them into open-source software and platforms.

Highlights 9 papers · 1 patent · 130+ citations (h-index 6) · 150+ GitHub stars · 20+ students supervised · 3 SNSF grants · 1 Innosuisse project · 2 industrial collaborations

Professional Experience

2025–now **Postdoc, University of Bern, Switzerland, Pattern Recognition Group.**

Structural spatio-temporal models for Swiss river-temperature forecasting (SNSF); a Transformer-plus-GNN extension accepted at **ICPR 2026**. Building LIULIAN, an open-source spatio-temporal data platform.

2024–now **Industry collaboration, N-Banker, Hong Kong, Digital Financial Services Research Center Ltd..**

I lead the AI strategy and the architecture and development of an LLM-agent system (chatbot, AI Studio proof-of-concept, and a data-fusion-platform design) for a neobank research center, with Hong Kong Polytechnic University; demonstrated at InnoEX 2026.

2024–2025 **Scientific Collaborator, HES-SO Fribourg (iCoSys, HEIA-FR), Switzerland.**

Graph and vision deep learning for engineering-diagram analysis (Innosuisse PLANALYSER) and handwritten historical documents; contributed to the SNSF Virtual Bodmer project.

2023–2024 **Research Fellow, University of Bern, Switzerland, Pattern Recognition Group.**

Graph-based machine-learning theory and applications; one paper at **ACPR 2023**.

2021–2022 **Postdoctoral Researcher, INSA Rouen, Normandie Université, France, COBRA lab.**

Graph machine learning for molecular redox-potential prediction (led to my **Journal of Computational Chemistry 2024** paper) and for polymer optimization (**OCTOPUSSY**).

2021 **Study Engineer, Université de Rouen, France.**

Education

2017–2021 **Ph.D. in Computer Science, INSA Rouen, Normandie Université, France, LITIS lab.**

Machine learning and pattern recognition in chemoinformatics: graph kernels, graph edit distances, and pre-image methods for graphs. Supervisors: Prof. Paul Honeine and Prof. Benoit Gaüzère.

2014–2017 **M.Sc. in Software Engineering, Xi'an Jiaotong University, China.**

Service-oriented control for software-defined networks; resulted in one China patent (CN106376041B).

2010–2014 **B.S. in Information Engineering, Xi'an Jiaotong University, China.**

2016 **Summer Program, National University of Singapore.**

Courses in biometrics, computational thinking, and community detection in large graphs (received A's).

Research Interests

Machine Learning on Graphs.

- Bridging graph spaces and methods: graph kernels, graph edit distances, and GNNs.
- Knowledge embedding in graphs.
- Generation and pre-image problems on graphs.

Applications of Machine Learning.

- Applying graph and spatio-temporal methods across computational chemistry, chemoinformatics, hydrology, engineering, and more.
- (Graph) machine learning, deep learning, and Large Language Models for computer vision, digital humanities, and finance.

Cross-Domain Collaboration

Environmental physics / hydrology	Spatio-temporal river water-temperature forecasting on a Swiss sensor network (SNSF; with Bern University of Applied Sciences); ICPR 2026.
Smart engineering	Innosuisse PLANALYSER: image-to-graph HVAC-drawing analysis (HES-SO Fribourg + WATTELSE AG).
Computational chemistry	Graph-based redox-potential prediction (J. Comp. Chem. 2024); OCTOPUSSY polymer optimization (COBRA + LITIS + University of Bath); the ECSC Open Round molecular-energy project (Interacting Quantum Atoms features).
Digital humanities	Graph and vision deep learning for handwritten historical documents and 3D papyrus reconstruction (SNSF Virtual Bodmer).
FinTech (research to industry)	N-Banker neobank platform: LLM-agent systems and data fusion (Digital Financial Services Research Center Ltd., with HK PolyU); demonstrated at InnoEX 2026.
Epidemiology (collaboration)	Epidemiology-informed deep networks for COVID-19 dynamics (Computers in Biology and Medicine 2023), as a co-author and collaborator.

Summary of Publications

Overview	9 papers and 1 patent · 130+ citations · h-index 6 on Google Scholar. Full list in the appendix.
Journals	5: J. Computational Chemistry 2024; Computers in Biology and Medicine 2023; Electronics 2022; Pattern Recognition Letters 2021; Expert Systems with Applications 2021.
Conferences and workshops	4: ICPR 2026 [Accepted]; ACPR 2023; and two S+SSPR 2020 workshop papers.
Patents	1: China Patent CN106376041B.

Research Grants and Projects

Ongoing

- 2025– **Spatio-temporal graph convolutional networks - a novel deep learning approach to forecasting river temperatures, SNSF 206352.**
- Collaboration: UniBE Pattern Recognition Group with Bern University of Applied Sciences.
 - My role: contributor and lead developer of the Transformer extension; accepted at ICPR 2026. [grant]

2024– **Combining Image and Graph-based Neural Networks for Handwriting Recognition, SNSF 217594.**

- Collaboration: iCoSys (HES-SO Fribourg) with TU Dortmund.
- My role: contributor. [grant]

In preparation

2026– **A fundable project proposal centred on LIULIAN, Innosuisse or a joint host.**

- Can be pursued as an Innosuisse Innovation Project (as Principal Investigator) or jointly with an industry or institutional host; the proposal can be brought along. [In preparation]

Completed

2024–2025 **PLANALYSER - Automated HVAC-Concept Audit and Optimisation using AI, Innosuisse.**

- Collaboration: iCoSys (HES-SO Fribourg) with WATTELSE AG.
- My role: one of the principal academic contributors. [grant]

2023–2024 **Novel State-of-the-Art Graph Matching Algorithms, SNSF 188496.**

- Collaboration: UniBE Pattern Recognition Group.
- My role: contributor; led to my first-author ACPR 2023 paper. [grant]

2023 **Open Round 2023/2, Support for Young Academics, UniBE ECSC.**

- Budget: CHF 3,100 (received).
- My role: **sole applicant** and host; a multi-scale graph-ML molecular-energy project.

2021–2023 **Original Computational Techniques for the Optimization of Polymers Using Sustainable SYNthesis (OCTOPUSSY), Academia (COBRA + LITIS + Univ. of Bath).**

- Graph machine learning applied to polymer optimization.
- My role: one of the key contributors.

2018–2021 **APi (Apprivoiser la Pré-image), ANR (PhD grant).**

- Topic: the pre-image problem for structured data.
- Output: three journal papers, two workshop papers, and the graphkit-learn library.

2014–2017 **Research on Service-oriented Programmable Control and Scheduling Scheme for Software Defined Network, Xi'an Jiaotong University (M.Sc.).**

- Extreme-learning-machine prediction of network mobility in named-data networks.
- Output: one China patent (CN106376041B).

Software and Open Source

Summary Several open-source libraries and platforms across graph machine learning, chemoinformatics, and spatio-temporal data; 150+ GitHub stars in total, with adoption by independent labs. Selected list in the appendix.

Teaching and Supervision

Student supervision ◦ **20+** BSc and MSc students supervised across the University of Bern, HES-SO Fribourg, and the N-Banker industry-academia partnership (predominantly Hong Kong, plus Switzerland).

- Topics: graph machine learning, computer vision, deep learning, and LLMs and agent systems.

Outreach talks Science-popularization and teaching-oriented talks for student and general audiences, for example at the Swiss-Chinese Science Club (2026) and a public lecture on AI tools (2026).

Recruitment and mentoring Served on the FinTech partnership recruitment panel; interviewed 20+ candidates from BSc to PhD level.

Invited Talks and Research Activities

- Jun 2024 **Invited speaker, GRAPHADON Summer School on Graphs for Data Analysis, Rouen, France.**
Talk: *Bridging Graph Spaces: Novel Algorithms and Their Applications*. [slides]
- Nov 2023 **Oral presentation, 7th Asian Conference on Pattern Recognition (ACPR), Kitakyushu, Japan.**
Paper [C23]. [pdf]
- Jan 2021 **Video presentations, S+SSPR 2020 Workshops.**
Papers [W21a] and [W21b]. [video1] [video2] [link]
- Sep 2020 **2020 CSC-Seminar (Germany and France).**
Talk on a graph pre-image method; awarded Excellent Scholar. [slides] [award]
- 2018 **Machine Learning Summer School (MLSS), Madrid, Spain.**
54-hour course; poster session. [poster]

Professional Service

- 2024– Member, Swiss Association for Pattern Recognition (SAPR).
- 2024 Member, Marie Curie Alumni Association (China Chapter); reviewer, ICPR 2024.
- 2022 Associated member, LITIS lab.

Skills

- Machine learning GNNs · graph kernels · graph edit distances · pre-image and generation · Transformers · spatio-temporal models · LLMs and agents · PyTorch · scikit-learn
- Programming Python · C++ · Cython · MATLAB
- Scientific tools Gaussian · RDKit · DeepChem · HPC and Linux clusters · Git · $\LaTeX$
- Languages Chinese (native) · English (fluent; working language of all my research) · French (B2; TCF 2017, 400/600)
- Hobbies Dancing, Hiking

Appendix

Publications (Full List)

- [C26] **Linlin Jia**, Benjamin Fankhauser, Vidushi Bigler, Kaspar Riesen. **Benchmarking Transformers on Spatio-Temporal River Water Temperature Modeling**. International Conference on Pattern Recognition (ICPR), 2026. [\[Accepted\]](#)
- [J24] **Linlin Jia**, Éric Brémond, Larissa Zaida, Benoit Gaüzère, Vincent Tognetti, Laurent Joubert. **Predicting Redox Potentials by Graph-Based Machine Learning Methods**. Journal of Computational Chemistry, 2024. [\[doi\]](#) [\[code\]](#)
- [C23] **Linlin Jia**, Xiao Ning, Benoit Gaüzère, Paul Honeine, Kaspar Riesen. **Bridging Distinct Spaces in Graph-based Machine Learning**. Asian Conference on Pattern Recognition (ACPR), 2023. [\[pdf\]](#) [\[code\]](#)
- [J23] Xiao Ning, **Linlin Jia**, Yongyue Wei, Xi-An Li, Feng Chen. **Epi-DNNs: Epidemiological Priors Informed Deep Neural Networks for Modeling COVID-19 Dynamics**. Computers in Biology and Medicine, 2023. [\[doi\]](#)
- [J22b] **Linlin Jia**, Vincent Tognetti, Laurent Joubert, Benoit Gaüzère, Paul Honeine. **A Study on the Stability of Graph Edit Distance Heuristics**. Electronics, 2022. [\[doi\]](#)
- [J22a] **Linlin Jia**, Benoit Gaüzère, Paul Honeine. **Graph Kernels Based on Linear Patterns: Theoretical and Experimental Comparisons**. Expert Systems with Applications, 2021. [\[doi\]](#) [\[code\]](#)
- [J21] **Linlin Jia**, Benoit Gaüzère, Paul Honeine. **graphkit-learn: A Python Library for Graph Kernels Based on Linear Patterns**. Pattern Recognition Letters, 2021. [\[doi\]](#) [\[code\]](#)
- [W21b] **Linlin Jia**, Benoit Gaüzère, Paul Honeine. **A Graph Pre-image Method Based on Graph Edit Distances**. IAPR Joint International Workshops S+SSPR, Springer LNCS, 2021. [\[preprint\]](#)
- [W21a] **Linlin Jia**, Benoit Gaüzère, Florian Yger, Paul Honeine. **A Metric Learning Approach to Graph Edit Costs for Regression**. IAPR Joint International Workshops S+SSPR, Springer LNCS, 2021. [\[preprint\]](#)
- [P16] Qu Hua, Zhao Jihong, Wu Jinkang, **Jia Linlin**, et al. **A Kind of Name Data Network Mobility Switching Method Predicted Using ELM**. China Patent CN106376041B. [\[patent\]](#)

Software (Selected List)

- 2018– **graphkit-learn**, *Open-source library*.
Graph kernels, graph edit distances, and the graph pre-image problem; 128+ stars; backs my PRL 2021 and ESWA 2021 first-author papers.
- 2024– **LIULIAN**, *Open-source platform*.
Spatio-temporal data platform under active development; LLM-driven interfaces and multimodal scientific graph benchmarks; Innosuisse Innovation Project in preparation.
- 2025–2026 **swiss-river-network-benchmark**, *ICPR 2026 code*.
Benchmarks for the ICPR 2026 spatio-temporal Transformer river-temperature paper; Swiss and Zurich datasets.
- 2024 **RedoxPrediction**, *J. Comp. Chem. 2024 code*.
Graph-based molecular redox-potential prediction pipeline released with the paper.

2023 **[ged-cost-learn-framework](#)**, *ACPR 2023 code*.

Data-driven learning of graph-edit-distance costs; backs my ACPR 2023 first-author paper.